

Supplementary Summer Examinations 2024

Course: B. Tech.

Branch: Computer & Allied

Semester : IV

Subject Code & Name: Computer Architecture & Organization (BTCOC304)

Max Marks: 60

Date: 06/07/2024

Duration: 3 Hr.

Instructions to the Students:

1. All Questions are Compulsory.
2. Draw neat diagram wherever necessary.
3. Figures to right indicates full marks
4. Assume suitable data wherever necessary and mention it clearly

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		
A) State the basic fundamental units of computers and give brief explanation for each with neat diagram.	(Synthesis)	06
B) Write a short note on i)PC ii)IR iii) MAR iv) MDR	(Remember)	06
C) Explain in detail Instruction Fetching and Execution	(Understand)	06
Q.2 Solve Any Two of the following.		
A) Explain types of Operand and Operations with suitable instruction of assembly language.	(Understand)	06
B) Describe and list Addressing Modes with examples	(Remember)	06
C) Explain RISC and CISC architecture.	(Understand)	06
Q. 3 Solve Any Two of the following.		
A) Multiply using a sequential circuit for binary multiplication i) 11 x 7 ii) 7 x 8	(Synthesis)	06
B) Write a Booth algorithm for multiplication with a suitable example.	(Understand)	06
C) Solve the following using Binary division restoring algorithm i) 8 / 2 ii) 11 / 5	(Remember)	06
Q.4 Solve Any Two of the following.		
A) Explain RAM & DRAM with neat diagram.	(Remember)	06
B) Write a short note i) Magnetic Disk ii) Magnetic Tape	(Understand)	06
C) Explain Cache Mapping – Set associative mapping with neat diagram	(Understand)	06
Q. 5 Solve Any Two of the following.		
A) Explain Microprogrammed control unit.	(Understand)	06
B) Write short note on i) I/O channels ii) Direct memory Access (DMA)	(Remember)	06
C) Explain Micro-instruction Sequencing.	(Understand)	06

*** End ***